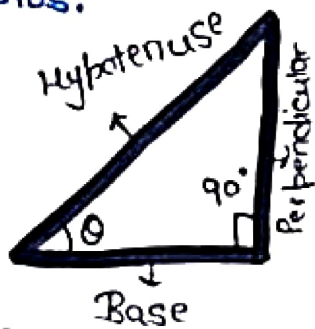


Basic Mathematics and calculus

* Trigonometry: Basic Trigonometry

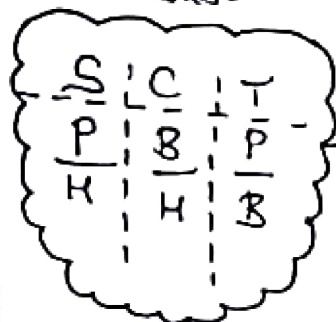
Ratios:



$$\sin \theta = \frac{P}{H}$$

$$\cos \theta = \frac{B}{H}$$

$$\tan \theta = \frac{P}{B} = \frac{\sin \theta}{\cos \theta}$$



Pandit Badri Prasad
Har Har Bhale

* Quotient Identities: $\tan \theta = \frac{\sin \theta}{\cos \theta}$

$$\text{ii) } \cot \theta = \frac{\cos \theta}{\sin \theta}$$

* Reciprocal Identities: $\cot \theta = \frac{1}{\tan \theta}$

$$\text{ii) } \operatorname{cosec} \theta = \frac{1}{\sin \theta} \quad \text{iii) } \sec \theta = \frac{1}{\cos \theta}$$

* Pythagorean Identities: i) $\sin^2 \theta + \cos^2 \theta = 1$

$$\text{ii) } \tan^2 \theta + 1 = \sec^2 \theta \quad \text{iii) } 1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$$

* Sum Identities Addition Formulas:

$$\text{i) } \sin(a+b) = \sin a \cos b + \cos a \sin b$$

$$\text{ii) } \cos(a+b) = \cos a \cos b - \sin a \sin b$$

$$\text{iii) } \tan(a+b) = \frac{\tan a + \tan b}{1 - \tan a \tan b}$$

* Difference Identities Subtraction

Formulas:

$$\text{i) } \sin(a-b) = \sin a \cos b - \cos a \sin b$$

$$\text{ii) } \cos(a-b) = \cos a \cos b + \sin a \sin b$$

$$\text{iii) } \tan(a-b) = \frac{\tan a - \tan b}{1 + \tan a \tan b}$$

* Half Angle Formulas:

$$\text{i) } \sin\left(\frac{\theta}{2}\right) = \pm \sqrt{\frac{1 - \cos \theta}{2}}$$

$$\text{ii) } \cos\left(\frac{\theta}{2}\right) = \pm \sqrt{\frac{1 + \cos \theta}{2}}$$

$$\text{iii) } \tan\left(\frac{\theta}{2}\right) = \frac{1 - \cos \theta}{\sin \theta} = \frac{\sin \theta}{1 + \cos \theta} = \pm \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}$$

* Sum to Product Formulas:

$$\text{i) } \sin a + \sin b = 2 \sin\left(\frac{a+b}{2}\right) \cos\left(\frac{a-b}{2}\right)$$

$$\text{ii) } \sin a - \sin b = 2 \sin\left(\frac{a-b}{2}\right) \cos\left(\frac{a+b}{2}\right)$$

$$\text{iii) } \cos a + \cos b = 2 \cos\left(\frac{a+b}{2}\right) \cos\left(\frac{a-b}{2}\right)$$

$$\text{iv) } \cos a - \cos b = 2 \sin\left(\frac{a+b}{2}\right) \sin\left(\frac{a-b}{2}\right)$$

* Product-to-Sum Formulas:

$$\text{i) } \sin a \sin b =$$

$$\frac{1}{2} [\cos(a-b) - \cos(a+b)]$$

$$\text{ii) } \sin a \cos b =$$

$$\frac{1}{2} [\sin(a+b) + \sin(a-b)]$$

Binomial Expression:

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots$$

• Applicable for small $x \ll 1$ (used in approximations like $\sqrt{1+x}$, $(1+x)^n$ etc).

Ex) $1! = 1$ $3! = 3 \times 2 \times 1 = 6$
 $2! = 2 \times 1$

Arithmetic Progression (AP):

• n th term: $a_n = a + (n-1)d$

• Sum of n terms: $S_n = \frac{n}{2}[2a + (n-1)d]$

Ex) AP: $a \rightarrow$ First term
 $d \rightarrow$ Difference

$$a_9 = 1 + (9-1)1$$

$$a_9 = 1 + 8 \times 1 = 9$$

$$S_{10} = \frac{10}{2}[2 + (10-1) \times 1] = 5[11] = 55$$

Geometric Progression (GP):

• n th term: $a_n = ar^{n-1}$

• Sum of n terms: $S_n = a \left(\frac{r^n - 1}{r - 1} \right)$, if $r \neq 1$

• Infinite GP Sum: $S = \frac{a}{1-r}$ (if $|r| < 1$)

Only valid for decreasing GP

Ex) G.P. = 2, 4, 8, 16, 32, 64, 128, ...
 $\downarrow$
 a Common Ratio = $r = 2$

12th term? | Sum till 5th term?

$$2 \left(\frac{2^5 - 1}{2 - 1} \right) = 62 \dots$$

Logarithms:

• Definition:

$$\log_b a = x = b^x = a$$

• Important Properties:

$$\log_a b = \log_a a + \log_a b, \log\left(\frac{a}{b}\right) = \log a - \log b, \log(a^n) = n \log a$$

• Base 10 (common log) and Base e (Natural Log $\ln$)

Basic Identities:

i) Log of a product:

$$\log_b(xy) = \log_b(x) + \log_b(y)$$

ii) Log of a quotient:

$$\log_b\left(\frac{x}{y}\right) = \log_b(x) - \log_b(y)$$

iii) Log of a power:

$$\log_b(x^n) = n \cdot \log_b(x)$$

iv) Log of 1:

$$\log_b(1) = 0$$

v) Log of the base itself:

$$\log_b(b) = 1$$

vi) Log of base raised to log:

$$\log_b(b^x) = x$$

Natural Logarithm (base e):

i) $\ln(e) = 1$

$$\ln\left(\frac{x}{y}\right) = \ln(x) - \ln(y)$$

ii) $\ln(1) = 0$

$$\ln(x^n) = n \cdot \ln(x)$$

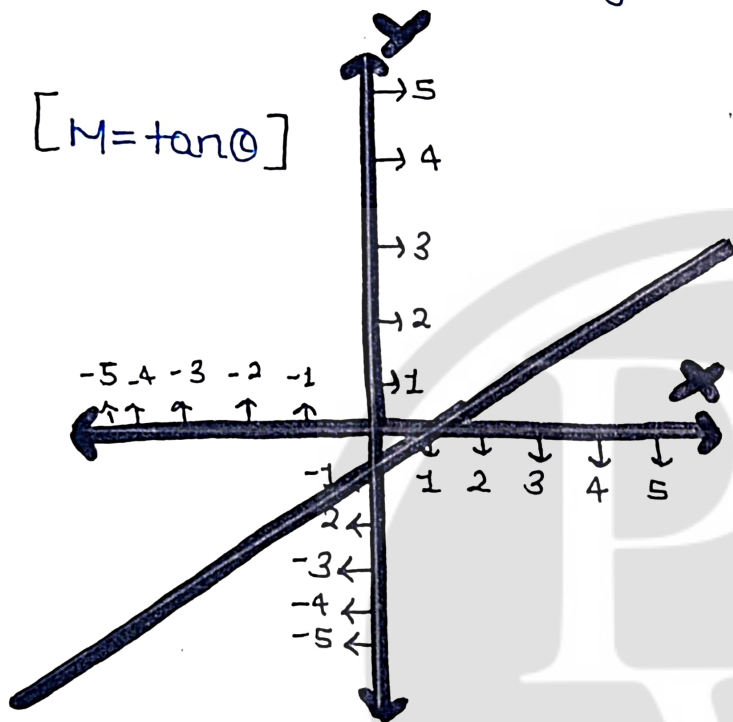
iii) $\ln(xy) = \ln(x) + \ln(y)$

Graphs: Graphs are visual tools to understand how two quantities are related.

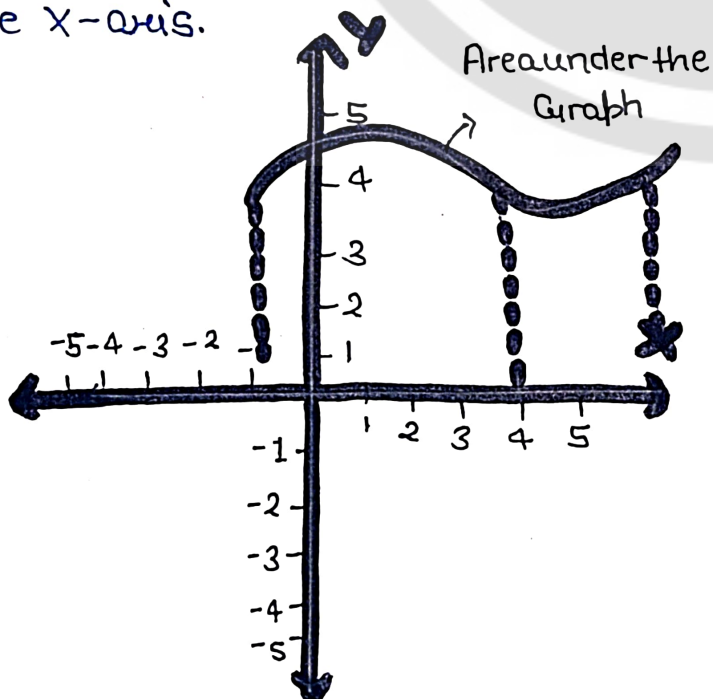
Commonly used in: motion, thermodynamics, etc.

Slope of Graph: denoted by 'm'

$$[M = \tan \theta]$$

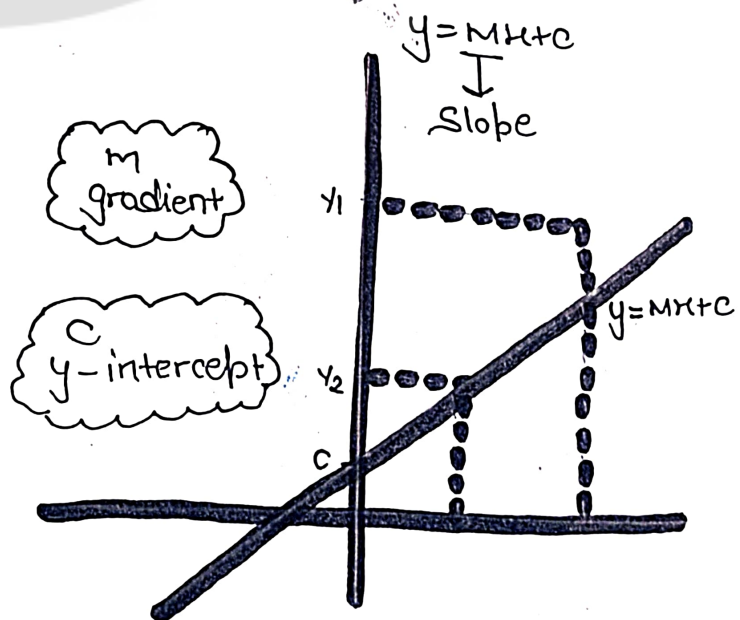
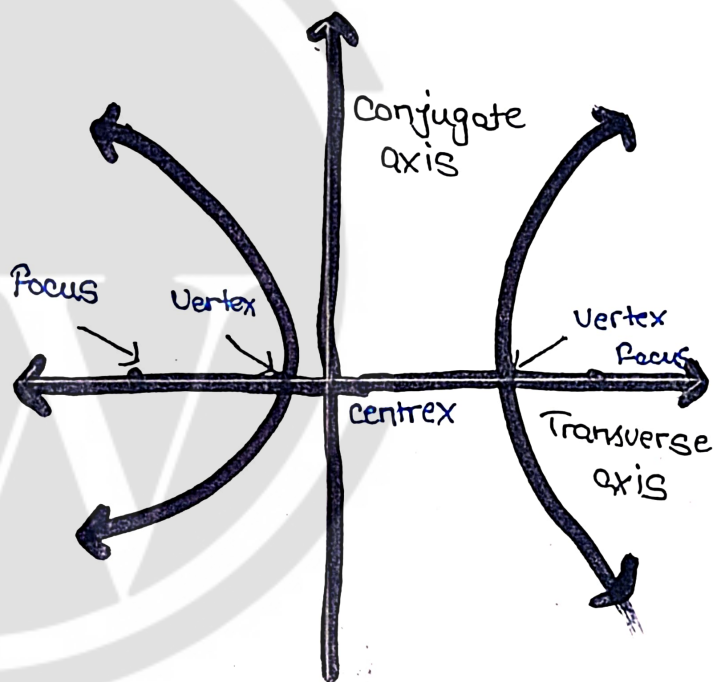
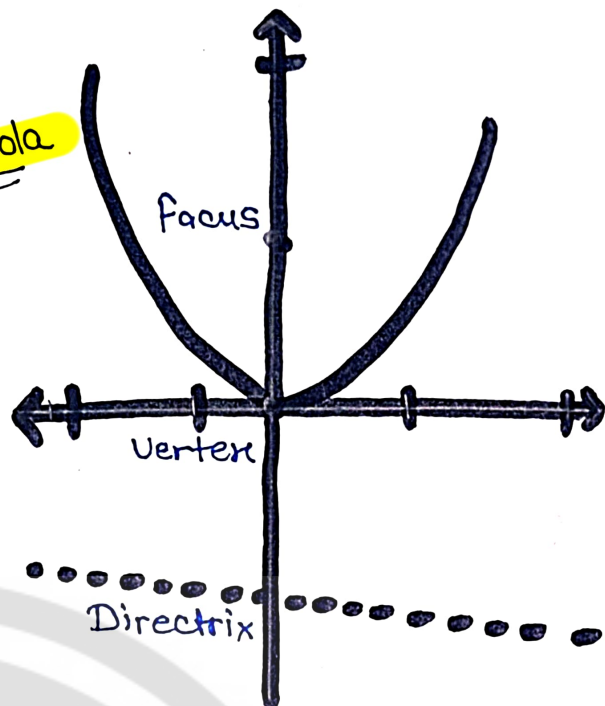


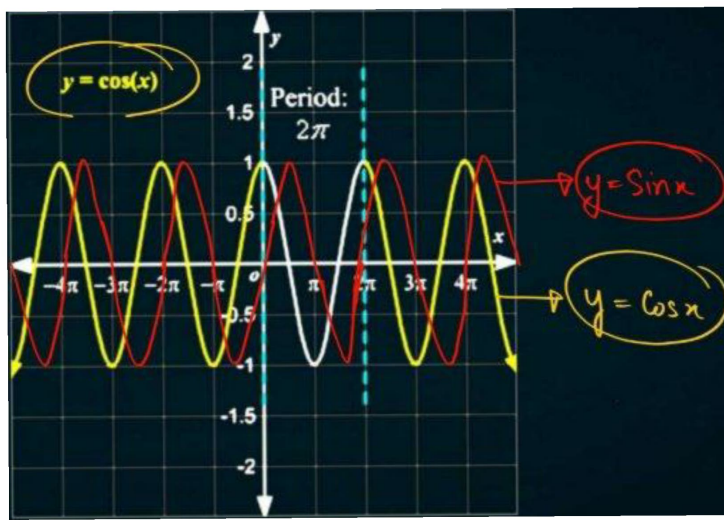
Area under the Graph: Area under the graph is the projection of Plot on the x-axis.



Some Important Graphs:

Parabola





Integration (Antiderivates)

→ Integration = Adding up small quantities to get the total.

→ In Physics

Quantity	Integration
Displacement	$s = \int v dt$
Velocity	$v = \int a dt$
Work done	$w = \int F dx$

→ Integration: Adding up small quantities to get the total

→ Note: C is the constant of integration.

→ Basic Formulas:

Function	Integral
→ $\int x^n dx$	→ $\frac{x^{n+1}}{n+1} + C, n \neq -1$
→ $\int \sin x dx$	→ $-\cos x + C$
→ $\int \cos x dx$	→ $\sin x + C$
→ $\int e^x dx$	→ $e^x + C$
→ $\int \frac{1}{x} dx$	→ $\ln x + C$

Differentiation (Derivatives):

Differentiation = Finding the rate of change

→ If $y = f(x)$, then $\frac{dy}{dx}$ tells show how y changes as x change.

Quantity	Derivative
Velocity	$v = \frac{ds}{dt}$
Acceleration	$a = \frac{dv}{dt} = \frac{d^2s}{dt^2}$
Force	$F = \frac{dU}{dx}$

→ Basic Rules: If $f(x) = x^n$, then: $\frac{d}{dx}(x^n) = nx^{n-1}$.

Function	Derivative
x^n	nx^{n-1}
→ $\sin x$	→ $\cos x$
→ $\cos x$	→ $-\sin x$
→ e^x	→ e^x
→ $\ln x$	→ $\frac{1}{x}$